

02 Exploratory Data Analysis

EP

2026-02-20

Analysis of the Grid Model Data

Following creation of the final grid, exploratory data analysis is carried out to determine the types of models to be used and illustrate the data.

```
setwd("~/CSCT_Pipeline")

library(targets)
library(sf)
library(dplyr)
library(tidyr)
library(purrr)
library(ggplot2)
library(patchwork)
library(GGally)
library(skimr)

grid_model_all <- tar_read(grid_model_all)

df <- grid_model_all %>%
  sf::st_drop_geometry()

predictors <- c("n_days", "dtm_elev", "avg_imperv", "rofsw_risk", "rofrs_risk",
               "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",
               "dist_to_river", "pct_greenpace")
```

Plot the Study Area for each city

The following plots show each cities boundary (in red) and the 1km² grid squares (black).

```
setwd("~/CSCT_Pipeline")
project_crs <- 27700

cities <- tibble::tibble(
  city_label = c("Bristol", "Gateshead", "Newcastle", "Exeter", "Nottingham"),
  city_name  = c(
    "City of Bristol (B)",
    "Gateshead District (B)",
    "Newcastle upon Tyne District (B)",
    "Exeter District (B)",
```

```

    "City of Nottingham (B)"),
  grid_file = c(
    "Data/grid_1km_Bristol.gpkg",
    "Data/grid_1km_Gateshead.gpkg",
    "Data/grid_1km_Newcastle.gpkg",
    "Data/grid_1km_Exeter.gpkg",
    "Data/grid_1km_Nottingham.gpkg"))

grids <- cities %>%
  mutate(grid = map(grid_file, ~ st_read(.x) %>% st_transform(project_crs)))

## Reading layer 'grid' from data source
## 'C:\Users\ellsp\OneDrive\Documents\CSCT_Pipeline\Data\grid_1km_Bristol.gpkg'
## using driver 'GPKG'
## Simple feature collection with 323 features and 14 fields
## Geometry type: POLYGON
## Dimension: XY
## Bounding box: xmin: 322292.8 ymin: 160864.6 xmax: 365292.8 ymax: 183864.6
## Projected CRS: OSGB36 / British National Grid
## Reading layer 'grid' from data source
## 'C:\Users\ellsp\OneDrive\Documents\CSCT_Pipeline\Data\grid_1km_Gateshead.gpkg'
## using driver 'GPKG'
## Simple feature collection with 187 features and 14 fields
## Geometry type: POLYGON
## Dimension: XY
## Bounding box: xmin: 409539.1 ymin: 553747.5 xmax: 431539.1 ymax: 565747.5
## Projected CRS: OSGB36 / British National Grid
## Reading layer 'grid' from data source
## 'C:\Users\ellsp\OneDrive\Documents\CSCT_Pipeline\Data\grid_1km_Newcastle.gpkg'
## using driver 'GPKG'
## Simple feature collection with 146 features and 14 fields
## Geometry type: POLYGON
## Dimension: XY
## Bounding box: xmin: 414452.1 ymin: 562772.4 xmax: 430452.1 ymax: 576772.4
## Projected CRS: OSGB36 / British National Grid
## Reading layer 'grid' from data source
## 'C:\Users\ellsp\OneDrive\Documents\CSCT_Pipeline\Data\grid_1km_Exeter.gpkg'
## using driver 'GPKG'
## Simple feature collection with 66 features and 14 fields
## Geometry type: POLYGON
## Dimension: XY
## Bounding box: xmin: 289277.6 ymin: 86966.9 xmax: 298277.6 ymax: 96966.9
## Projected CRS: OSGB36 / British National Grid
## Reading layer 'grid' from data source
## 'C:\Users\ellsp\OneDrive\Documents\CSCT_Pipeline\Data\grid_1km_Nottingham.gpkg'
## using driver 'GPKG'
## Simple feature collection with 106 features and 14 fields
## Geometry type: POLYGON
## Dimension: XY
## Bounding box: xmin: 450661.2 ymin: 332708.2 xmax: 461661.2 ymax: 347708.2
## Projected CRS: OSGB36 / British National Grid

```

```
grid_model_all <- map2(grids$grid, grids$city_label, ~ mutate(.x, city_label = .y)) %>%
  bind_rows()
```

```
boundary <- st_read("Data/OS_borders/district_borough_unitary_region.shp") %>%
  st_transform(project_crs)
```

```
## Reading layer 'district_borough_unitary_region' from data source
##   'C:\Users\ellsp\OneDrive\Documents\CSCT_Pipeline\Data\OS_borders\district_borough_unitary_region.shp'
##   using driver 'ESRI Shapefile'
## Simple feature collection with 350 features and 15 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: 5512.998 ymin: 5333.81 xmax: 655992.8 ymax: 1220310
## Projected CRS: OSGB36 / British National Grid
```

```
boundaries <- boundary %>%
  inner_join(cities %>% select(city_name, city_label), by = c("NAME" = "city_name"))
```

```
grid_cities <- st_within(grid_model_all, boundaries %>% select(city_label))
```

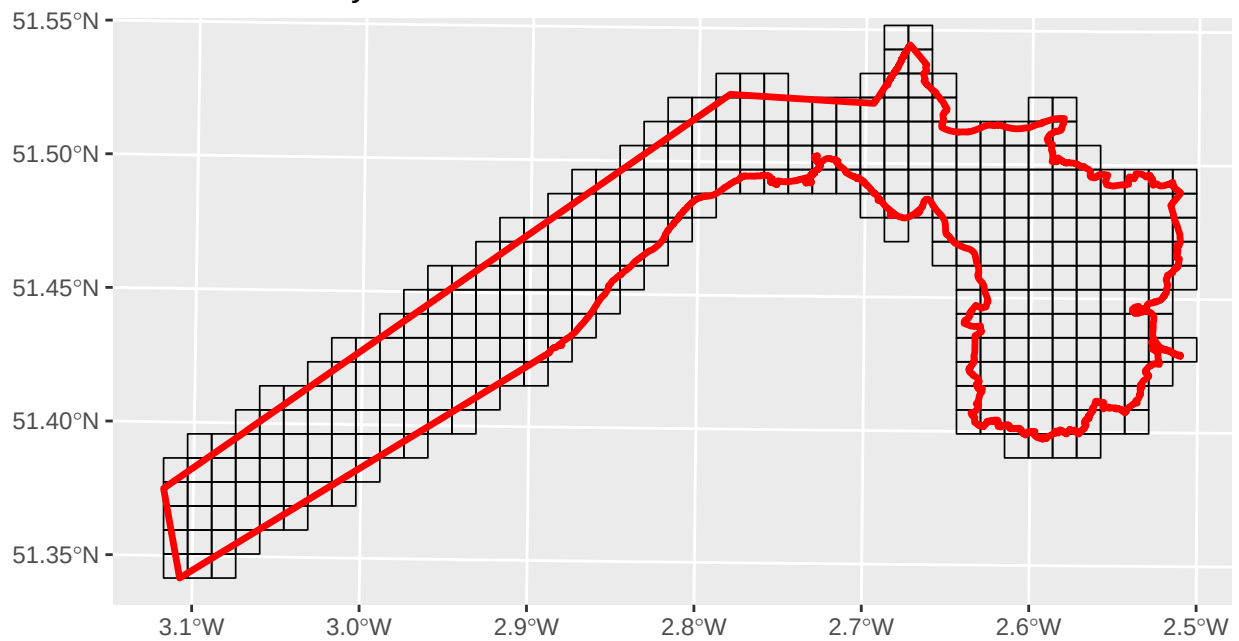
```
plot_grid <- function(city){
  city_grid      <- grid_model_all %>% filter(city_label == city)
  city_boundary <- boundaries %>% filter(city_label == city)
```

```
ggplot() +
  geom_sf(data = city_grid, colour = "black", fill = NA, linewidth = 0.3) +
  geom_sf(data = city_boundary, colour = "red", fill = NA, linewidth = 1.2) +
  coord_sf(
    xlim = st_bbox(city_boundary)[c("xmin", "xmax")],
    ylim = st_bbox(city_boundary)[c("ymin", "ymax")],
    expand = TRUE) +
  labs(title = paste(city, "Boundary with Grid"), x = NULL, y = NULL) +
  theme_gray() }
```

```
plots <- set_names(cities$city_label) %>%
  map(plot_grid)
```

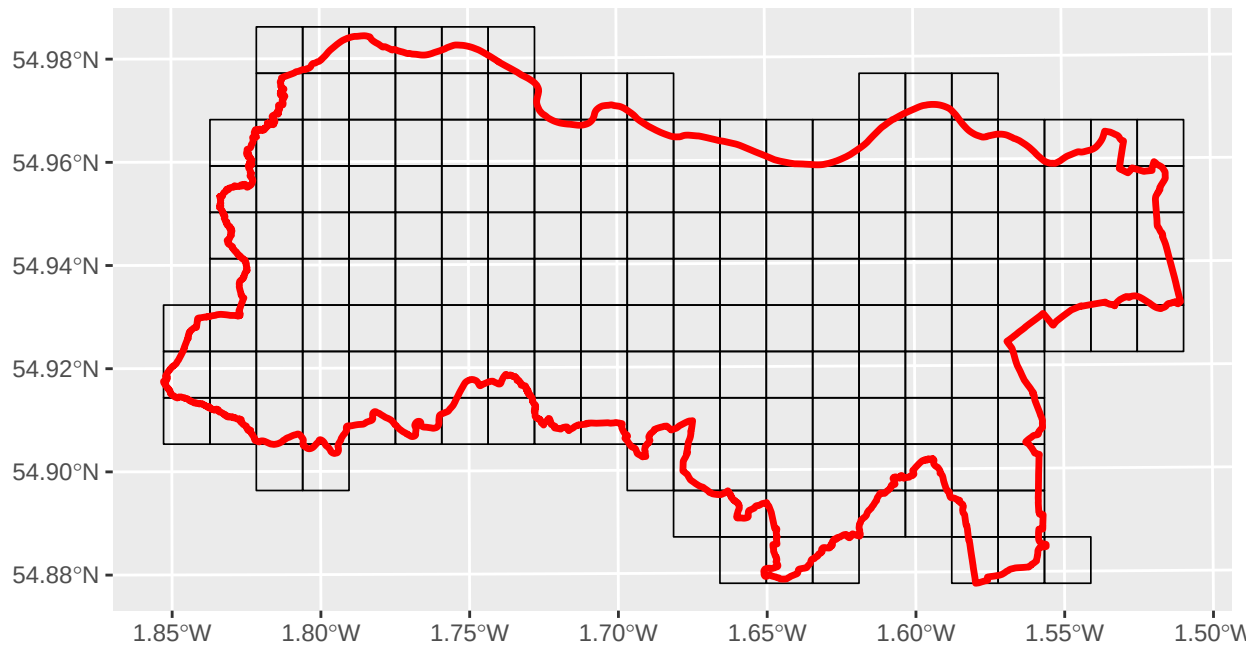
```
plots$Bristol
```

Bristol Boundary with Grid



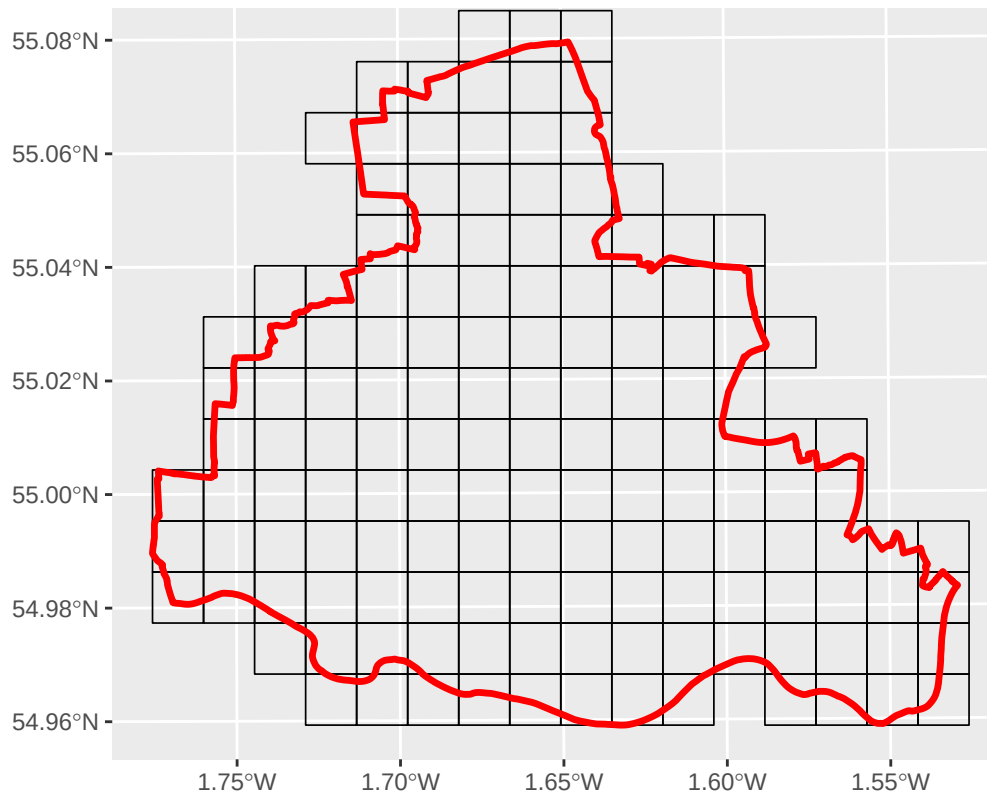
plots\$Gateshead

Gateshead Boundary with Grid



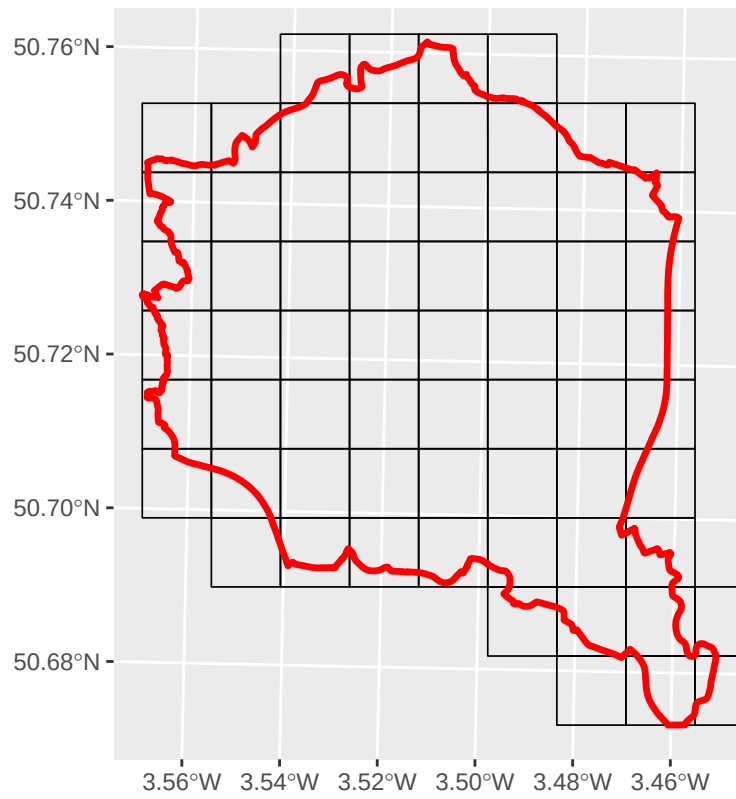
plots\$Newcastle

Newcastle Boundary with Grid



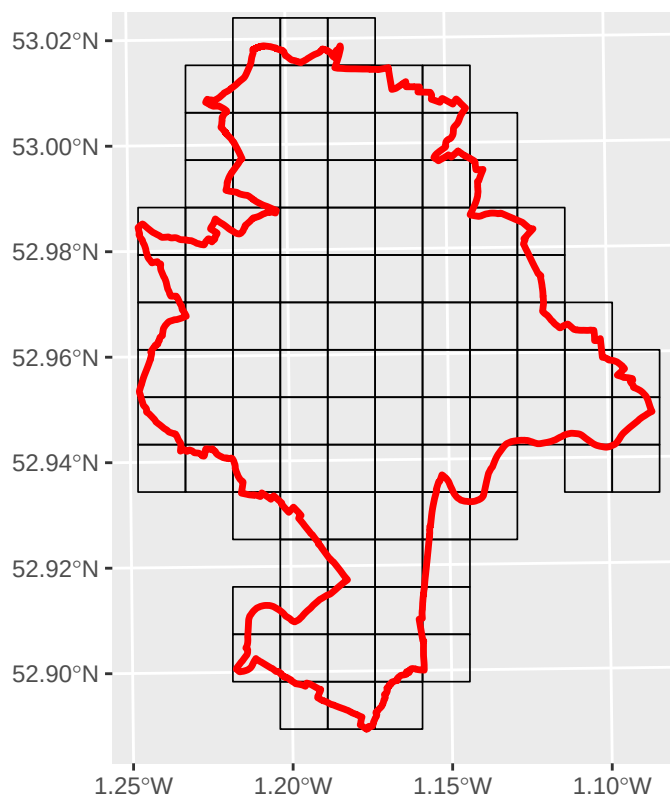
plots\$Exeter

Exeter Boundary with Grid



plots\$Nottingham

Nottingham Boundary with Grid

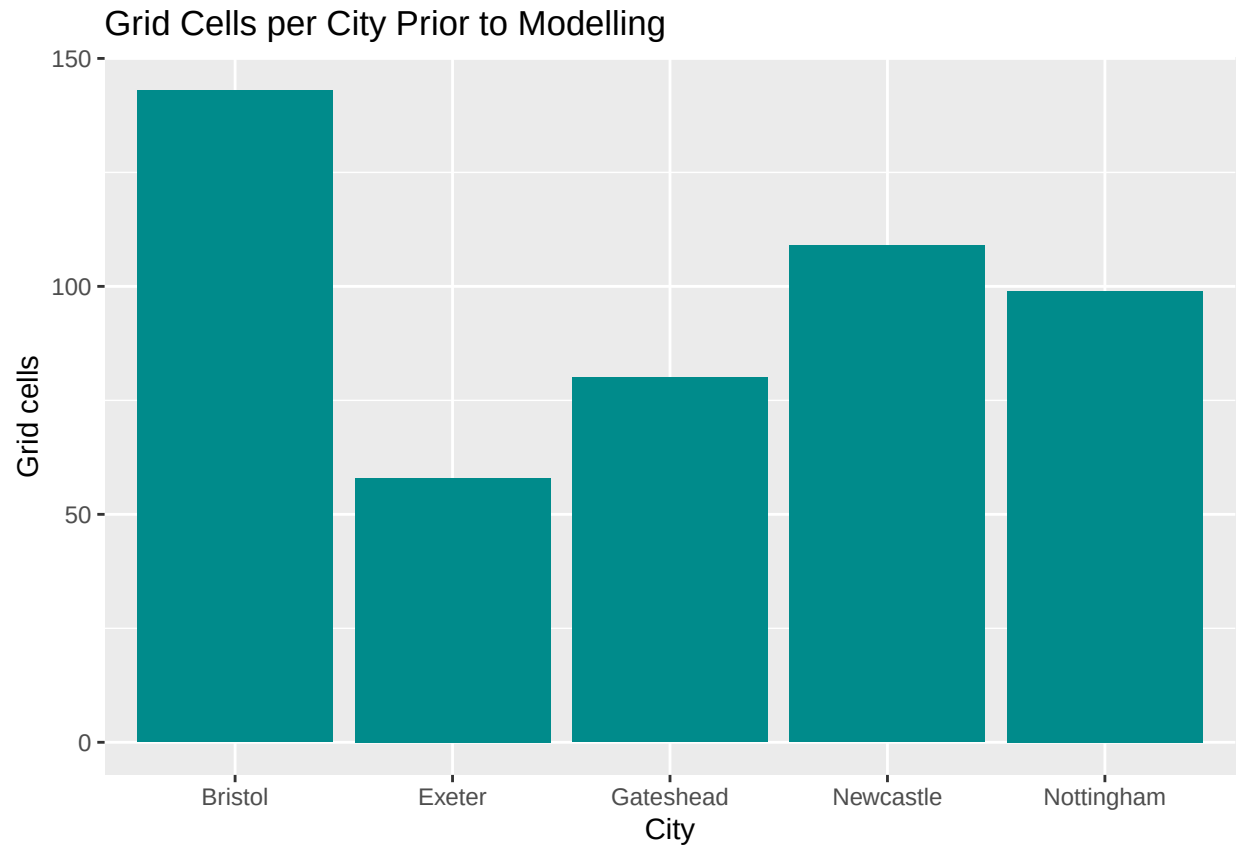


```
#ggsave("C:/Users/ellsp/OneDrive/Documents/1 Master's Project/Graphics/gateshead_grid.png", plot = plot)
#ggsave("C:/Users/ellsp/OneDrive/Documents/1 Master's Project/Graphics/newcastle_grid.png", plot = plot)
#ggsave("C:/Users/ellsp/OneDrive/Documents/1 Master's Project/Graphics/exeter_grid.png", plot = plots$E)
#ggsave("C:/Users/ellsp/OneDrive/Documents/1 Master's Project/Graphics/Nottingham_grid.png", plot = plot)
```

Number of grid cells

These plots show the number of grid cells per city, prior to modelling. This is the total number of grid squares, before rows are removed due to missingness.

```
cityplot<- df %>%
  count(city) %>%
  ggplot(aes(city, n)) +
  geom_col(fill = "cyan4") +
  labs(x = "City", y = "Grid cells", title = "Grid Cells per City Prior to Modelling") +
  theme_grey()
cityplot
```



```
#ggsave("../Graphics/Grid_Cells_per_City.png", cityplot, width = 10, height = 6, dpi = 1000 )
```

Numerical Summary Statistics

These summary statistics look at the numerical variables in the grid.

```
numerical_predictors <- c("n_days", "dtm_elev", "avg_imperv",
                          "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",
                          "dist_to_river", "pct_greenpace")

summary_table <- df %>%
  select(all_of(numerical_predictors)) %>%
  pivot_longer(
    cols = everything(),
    names_to = "Predictor",
    values_to = "Value") %>%
  group_by(Predictor) %>%
  summarise(
    Mean = mean(Value, na.rm = TRUE),
    SD = sd(Value, na.rm = TRUE),
    Min = min(Value, na.rm = TRUE),
    Max = max(Value, na.rm = TRUE),
    .groups = "drop")

summary_table
```

```
## # A tibble: 9 x 5
##   Predictor      Mean      SD      Min      Max
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
## 1 avg_area_ha      4.80    10.8      0      181.
## 2 avg_imperv      36.2    19.1    0.435    84.3
## 3 dist_to_river    718.    605.     1.18   2994.
## 4 dtm_elev        51.4    30.9     1.06    152.
## 5 n_days          386.    594.      1    3955
## 6 pct_greenpace     0.126    0.160     0       1.27
## 7 treeht           5.01     1.46     0      10.5
## 8 weighted_precip  771.     95.5    639.    1017.
## 9 weighted_temp    10.3     0.616    9.19    11.1
```

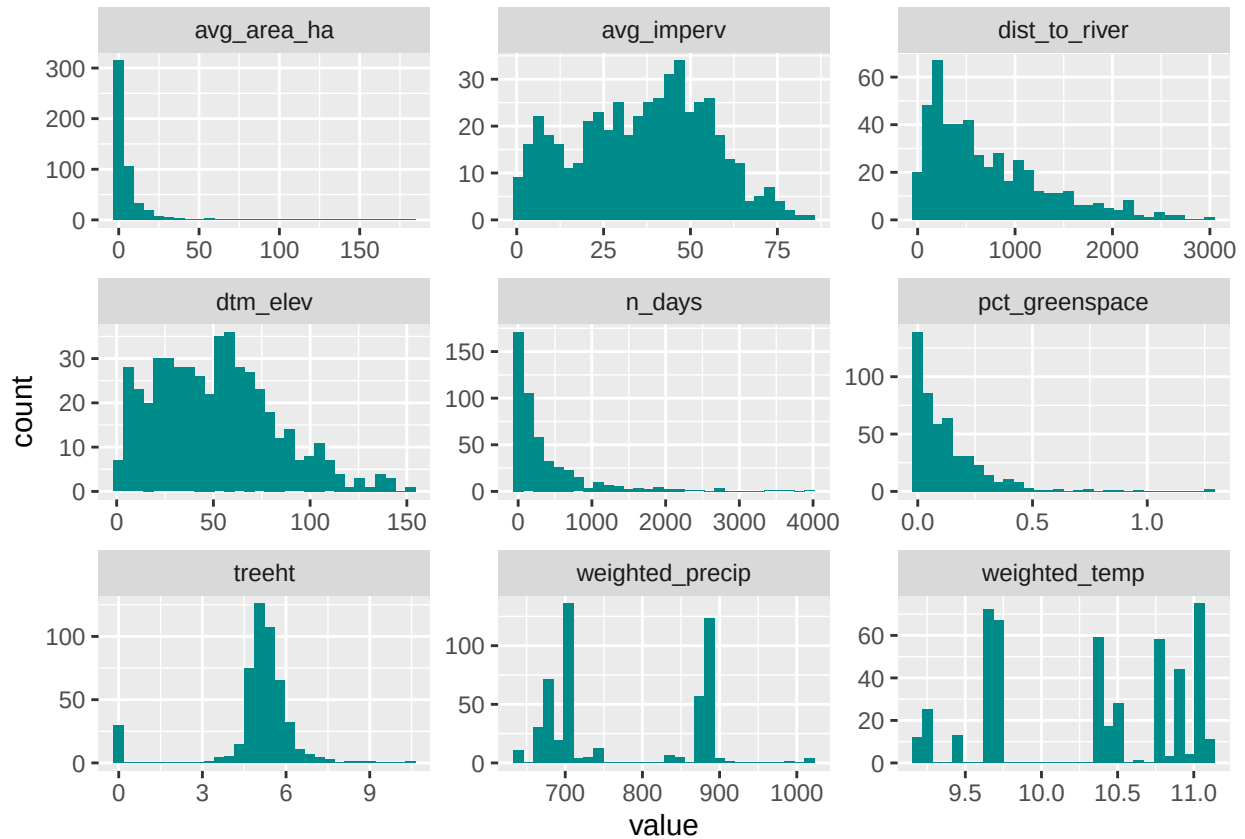
Distributions

The distribution of these numerical variables is varied. There is some skew in the data and multiple peaks in the the precipitation and temperature categories. This is to be expected since the cities are geographically distinct from each other.

```
dist_hists <- df %>%
  pivot_longer(all_of(numerical_predictors)) %>%
  ggplot(aes(value)) +
  geom_histogram(bins = 30, fill = "cyan4") +
  facet_wrap(~name, scales = "free")
```

```
dist_hists
```

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## ('stat_bin()').
```



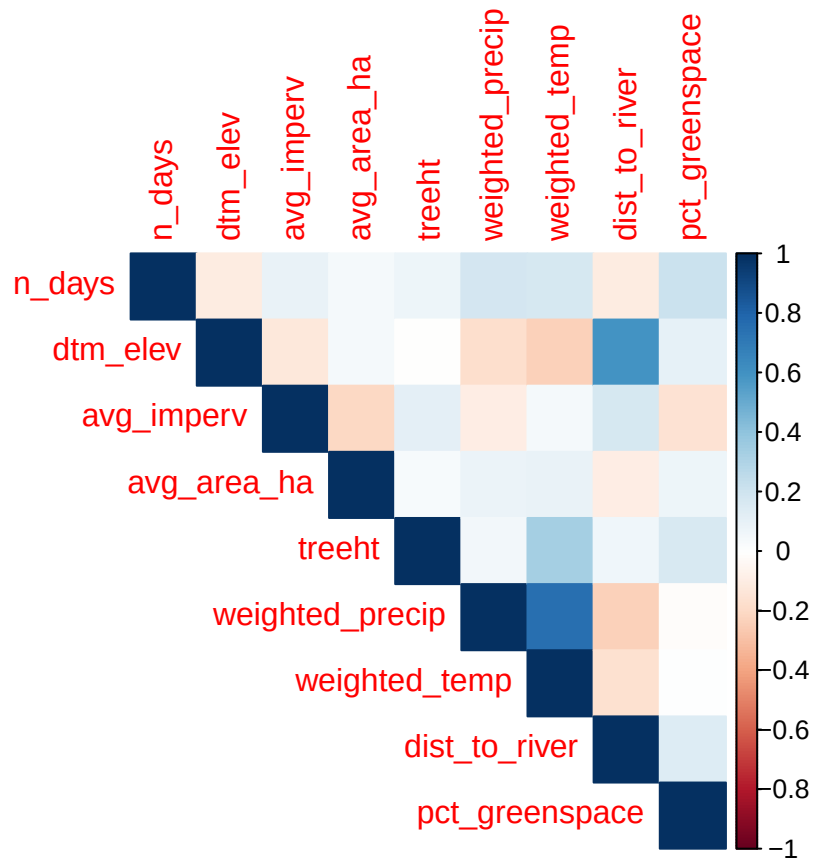
```
#ggsave("../Graphics/Distribution_Histograms.png", dist_hists, width = 10, height = 6, dpi = 1000 )
```

Correlation Matrix

Correlation analysis indicates that there is very little correlation between variables. The strongest correlations lie between distance to river and elevation, and precipitation and temperature. These are both environmentally logical.

```
library(corrplot)

corrplot(
  cor(df[numerical_predictors], use = "pairwise.complete.obs"),
  method = "color",
  type = "upper")
```



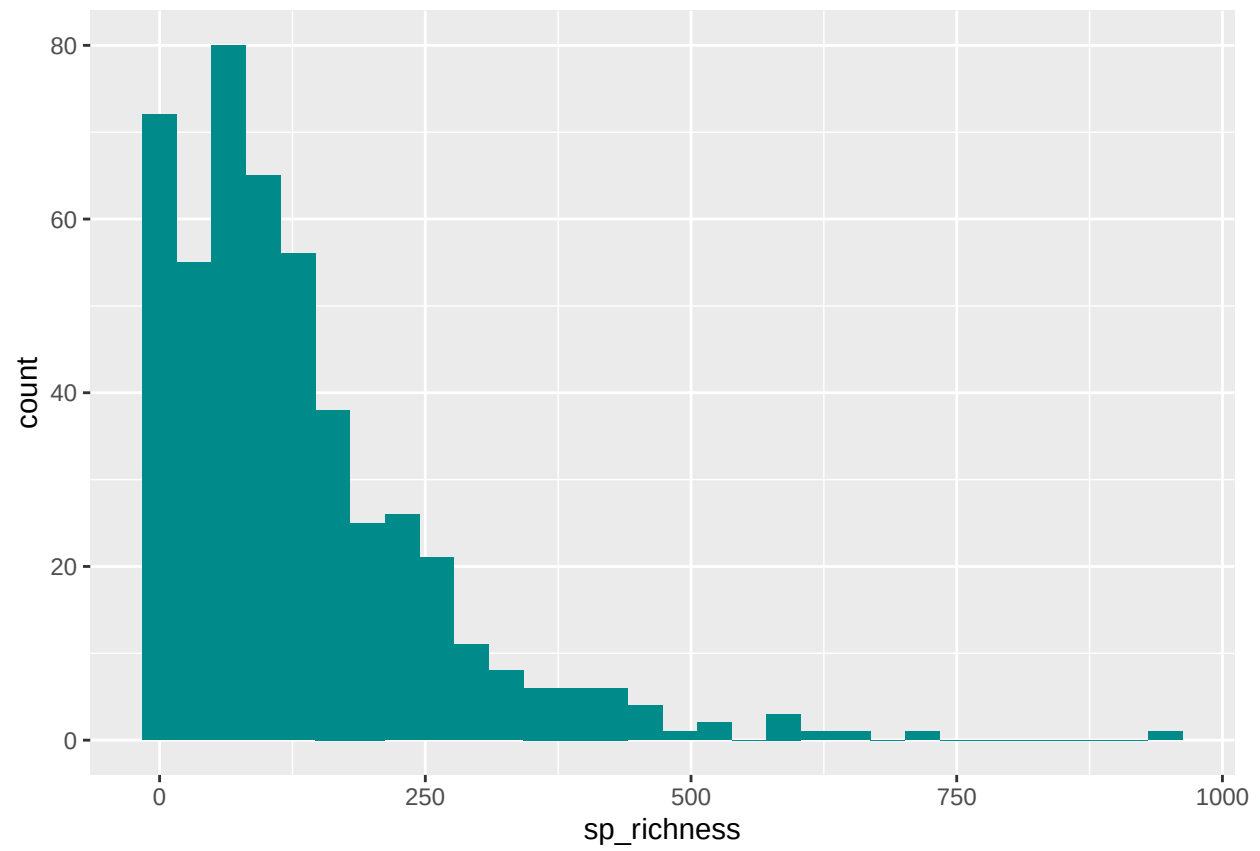
Richness and Simpson

These histograms show non-normally distributed data. The richness data is heavily skewed and overdispersed, with a few extreme outliers. The Simpson histogram also shows highly skewed data, bound between 0 and 1, but with a small, long tail. To account for this, a negative binomial regression will be used for the richness diversity indice and a beta regression model for the Simpson diversity indice.

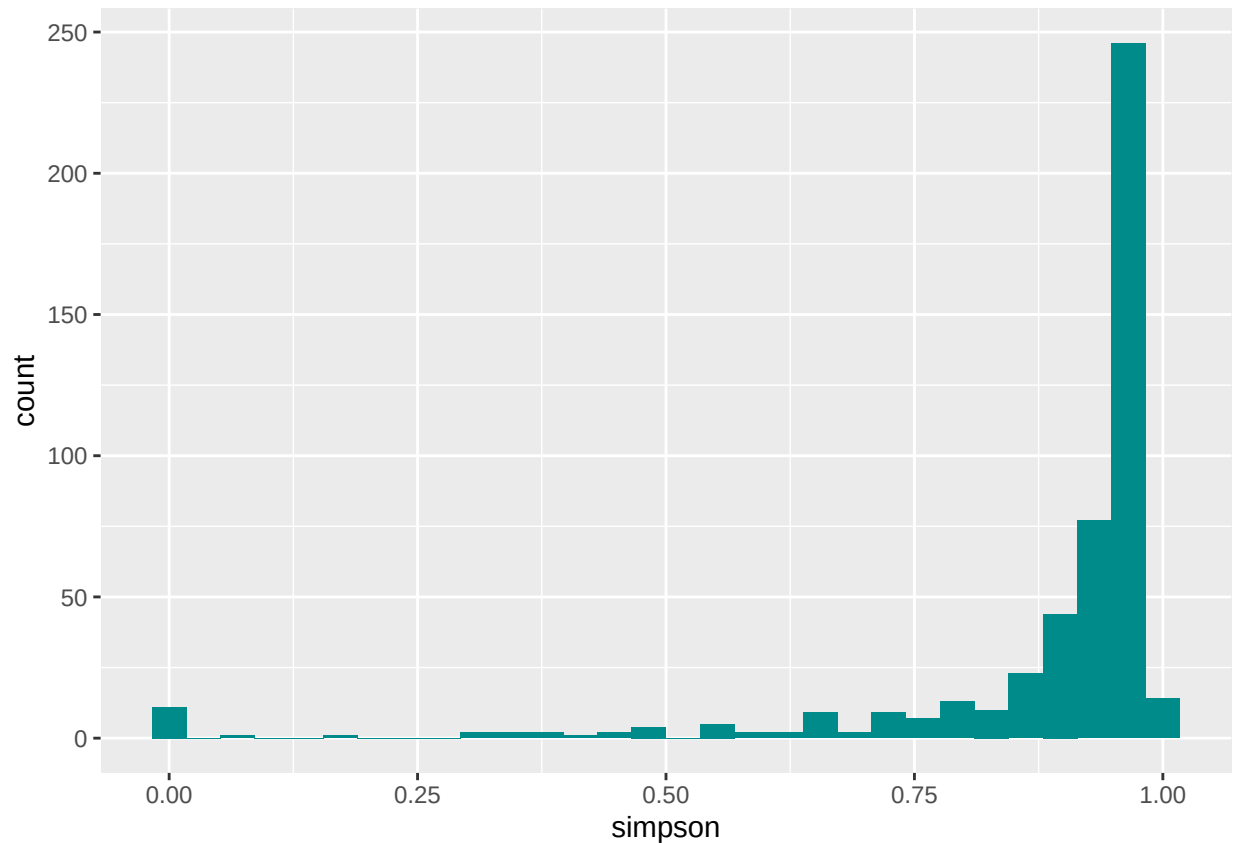
```
rich_hist <- ggplot(df, aes(sp_richness)) +
  geom_histogram(fill = "cyan4")

simp_hist <- ggplot(df, aes(simpson)) +
  geom_histogram(fill = "cyan4")

rich_hist
```



simp_hist



```
#ggsave("../Graphics/Richness_Histogram.png", rich_hist, width = 10, height = 6, dpi = 1000 )
#ggsave("../Graphics/Simpson_Histogram.png", simp_hist, width = 10, height = 6, dpi = 1000 )
```

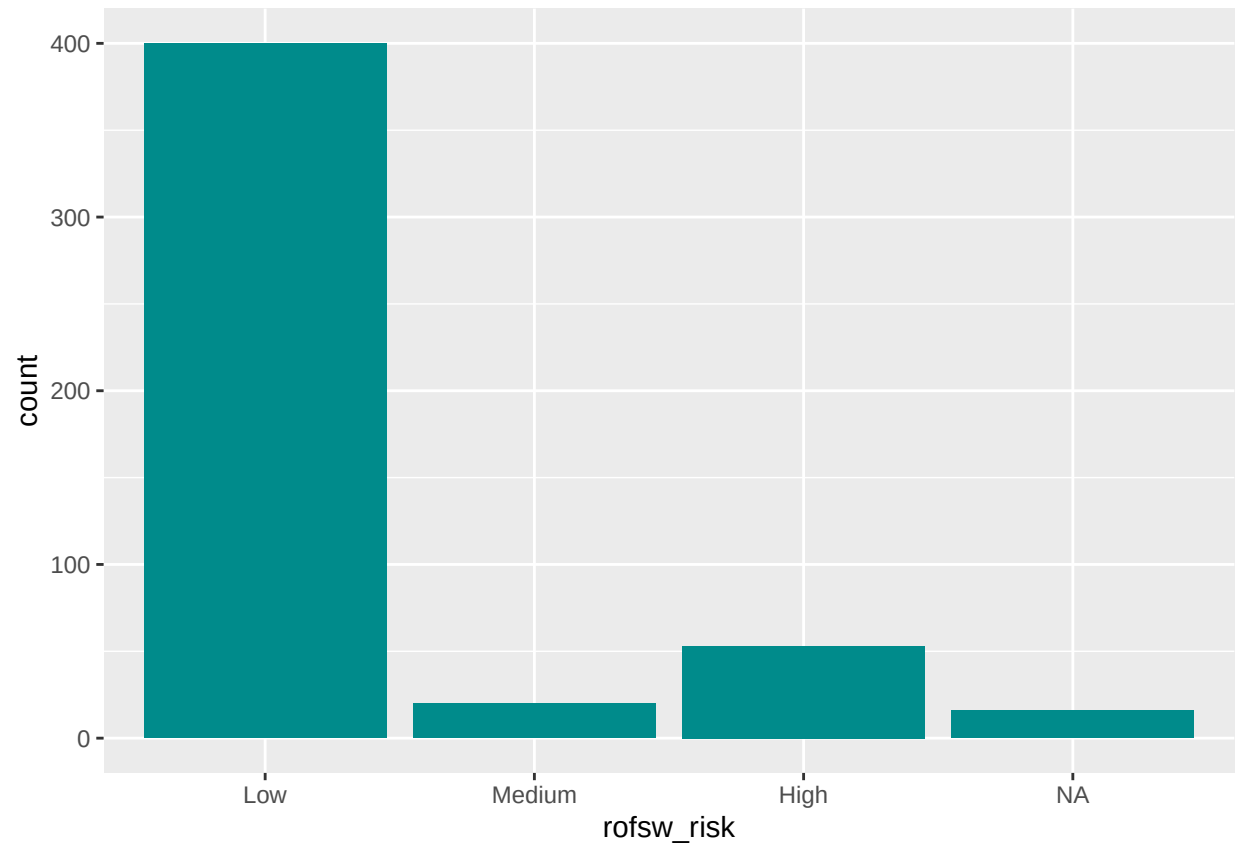
ROFSW and ROFRS

These bar plots show the spread of categorical variables, Risk of flooding from surface water (ROFSW) and Risk of flooding from rivers and seas (ROFRS). The variance between categories is large in both variables, but especially the ROFSW. In ROFRS, there is a huge number of grid cells with no data associated. This may prove an unreliable predictor when modelling biodiversity.

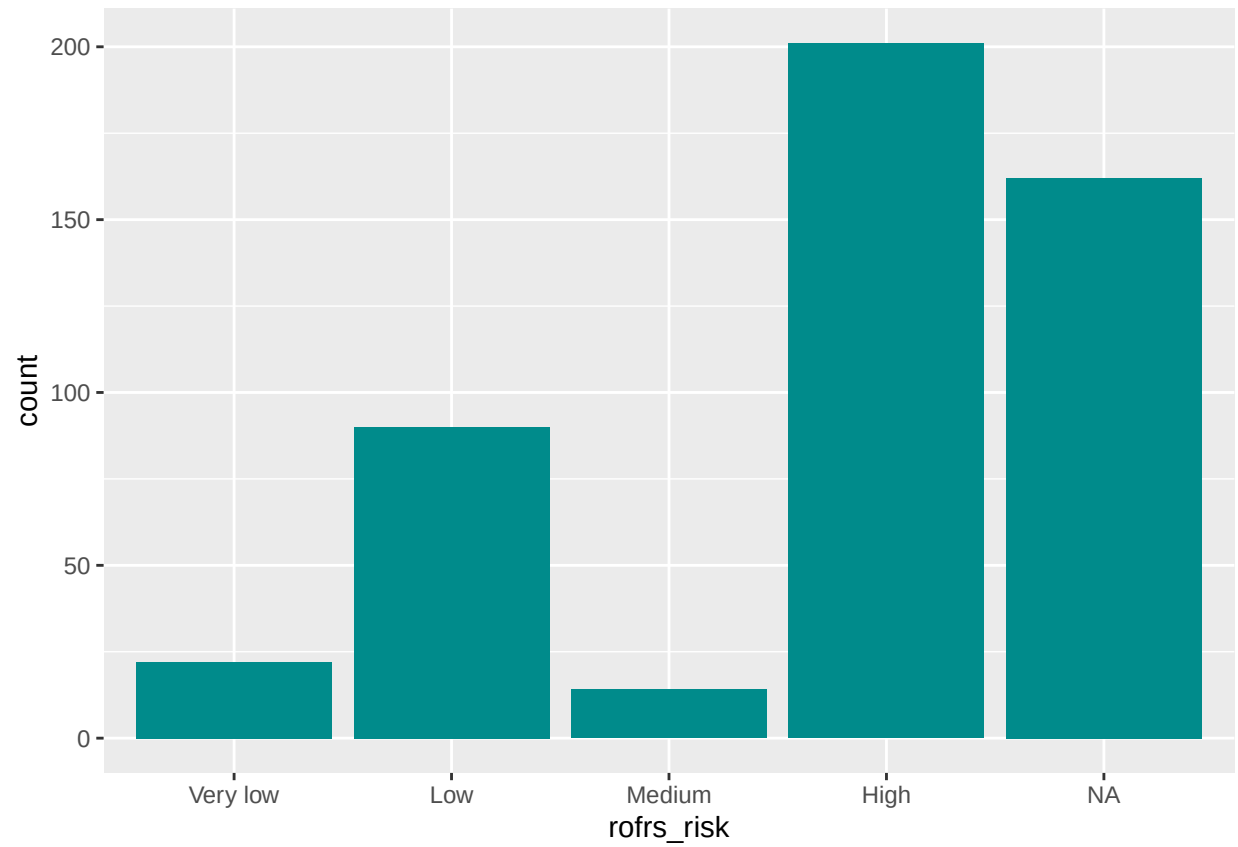
```
rofsw <- ggplot(df, aes(rofsw_risk)) +
  geom_bar(fill = "cyan4")

rofrs <- ggplot(df, aes(rofrs_risk)) +
  geom_bar(fill = "cyan4")

rofsw
```



rofrs



```
#ggsave("../Graphics/ROFSW_Bar.png", rofsw, width = 10, height = 6, dpi = 1000 )  
#ggsave("../Graphics/ROFRS_Bar.png", rofrs, width = 10, height = 6, dpi = 1000 )
```